BOOVARAGAVAN R V

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

- Electronics and Communication Engineering student with skills in Programming, Embedded Systems and Hardware Prototyping.
- Hands on Experience with Raspberry Pi, Arudino, ESP32, NodeMCU and Developing Automation Project for Real World Applications
- Skilled in branding, team building and management with demonstrated success
- Strong learner with good team work, communication and problems solving skills, seeking opportunities to gain industry experience

SKILLS

Programming Languages: Java, Python, SQL
Databases: MySQL
Soft Skills: Critical Thinking, Leadership, Problem-Solving
Tools & Technologies: Git, VS Code, Figma, Canva, TinkerCad, Claude
Technical Skills: Laser Source Servicing, Optical fiber splicing, Fiber Cleaving, Power Testing and Fiber Cable Replacing

EDUCATION

Bachelor of Engineering — Chennai, India
Sathyabama Institute of Science and Technology — 2023-2027
***Relevant Course Work**: Software Development* — **CGPA: 7.8/10 (Till 6th Semester)**

Higher Secondary Education — Namakkal, India
Government Higher Secondary School, Konur — 2023 - **Percentage: 82%**

CERTIFICATIONS

- Google UI / UX
- Google Generative AI
- HackerRank Basic Java
- NPTEL – Introduction to Internet of Things
- NPTEL – Sensor Technologies: Physics, Fabrication and Circuits.

PROJECTS

Automated Number Plate Recognition using Raspberry Pi — **September 2024**
Python, Open CV, Excel Automation

• Designed a system automatically detect college bus arrival and departure using number plate recognition
• Captures live images using a camera module, identifies bus numbers, and records entry/exit time without manual effort. Automatically stores all bus logs in an Excel sheet, making data easily accessible for security staff and administration.

Health Chatbot for Disease Awareness — **October 2025**
Python, Twilio

- Built a health-oriented chatbot using Twilio to deliver disease information and awareness through automated response.
- Designed the chatbot for easy user interaction and quick access to basic health-related details.
- Aimed to improve health awareness and encourage early understanding of symptoms.

VOTE GUARD – Smart Voter Verification **January 2026**

Tech Stack: Python, Streamlit, Pandas, RapidFuzz, ESP32, RC522 RFID, Arudino IDE

- Developed an AI-assisted voter verification system to detect ghost and inconsistent voter records.
- Implemented rule-based risk analysis for classifying records into Low, Medium, and High Risk levels.
- Built an interactive Streamlit dashboard with filtering and downloadable reports.
- Integrated ESP32 and RC522 RFID module for voter identify verification prototype.
- Applied similarity analysis for detecting suspicious voter data patterns.

Adaptive Indoor Air Quality Control System **April 2026**

Tech Stack: Python, Arudion IDE, ESP32, NodeMCU, MQ135, DHT11, IoT, Serial Communication

- Developed a smart indoor air quality monitoring and adaptive ventilation control system using embedded system.
- Implemented real-time monitoring of temperature, humidity, and harmful gas levels using MQ135 and DHT11.
- Designed adaptive fan and ventilation control logic based on indoor air quality conditions.
- Built a real-time dashboard for live sensor monitoring, graphical visualizations and alert notifications
- Integrated ESP32 / NodeMCU with IoT platforms for remote monitoring and automation.
- Implemented serial communication, data logging and automatic alert systems for poor air quality detection.

EXPERIENCE

Software Development Internship **India**

New Tech Solutions, Salem **July 2025**

- Worked on live software development projects under the guidance of experience mentors
- Assisted in designing, developing, testing and debugging software applications.
- Collaborated with the development team to implement project requirements and improve application functionality
- Gained hands-on-experience with software development practices, version control, and debugging tech

Repair and Service Internship **India**

Laser Experts India LLP – Hosur **September 2026**

- Performed Optical Fiber Cable replacement, Cleaving, Splicing, Source assembly and Power Testing under supervision
- Gained exposure to Laser source alarms, diode and power module inspections, QBH components and Chiller maintenance.
- Participated in customer site repairs, Laser Source inspections and cutting trials.
- Managed Customer records, Invoices, delivery Challans and Consumables dispatch using Zoho Books
- Assisted in repairing and troubleshooting IPG, Max photonics and other industrial laser Source machines

POSITION OF RESPONSIBILITY

Microsoft Learn Student Ambassador **India**

Microsoft **July 2025**

- Conducted workshops and mentoring sessions on AI-powered development tools like GitHub Copilot and Azure AI.
- Collaborated with global student communities to build open-source projects and promote Microsoft technologies.
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Design Team Lead and Organizing Lead **India**

ISTE Student Chapter, Sathyabama University **August 2023**

- Successfully conducted a 2 Electronics related Workshops and 3 Technical Events
- Creating Attractive Brochures and Posters for Conduct an Event

Google Student Ambassador

Google **India**

October 2025

- Coordinate events, workshops, and awareness drives focused on Google Technologies and developer tools
- Improved communication, leadership, and technical collaboration skills while contributing to growing developer community